NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2022-2023

IE2104 – DIGITAL ELECTRONICS

April / May 2023 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 5 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. (a) Derive the minterm expression of the following logic function in the form of $y(a,b,c,d) = \Sigma m(\ldots)$.

$$y(a,b,c,d) = \overline{(\bar{c} + a \bullet d)} \oplus (\bar{b} \bullet c + \bar{a})$$

(10 Marks)

(b) Draw a CMOS circuit for the logic function $z(a,b,c,d,e) = (b+e) \bullet (c + d \bullet \bar{a})$ with minimum number of transistors.

(10 Marks)

(c) The conversion from 3-bit binary to Gray code is given in Table 1 on Page 2. Implement the conversion with **ONLY** the encoder and the decoder given in Tables 2 and 3 on Page 2 respectively. Label clearly the inputs and outputs.

(5 Marks)

Note: Question No. 1 continues page 2.

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Table 1

Binary			Gray code		
b_2	b_1	b_0	g_2	g_1	g_0
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	1	1	1
1	1	0	1	0	1
1	1	1	1	0	0

Table 2

Input								Output		
I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

Table 3

Input			Output							
I_2	I_1	I_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

2. (a) The people on Planet-S uses base-x number system. To them, the arithmetic of $316 + 28 = 341$ is correct. What is the base of the number system that the people on Planet-S use?

(5 Marks)

Note: Question No. 2 continues page 3.

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(b) Derive the minimized Product of Sums (POS) expression of the logic function $f(a,b,c,d) = \sum m(2,5,10,14,15) + x(0,7,8,12)$, where $x(0,7,8,12)$ are the don't-care terms.

(10 Marks)

(c) Given the logic function $g(a,b,c) = \Pi M(1,3,4,5)$ and the MUX in Table 4, and assuming the complement signals of a, b and c are NOT available, implement the function $g(a,b,c)$ with **ONLY** the MUX. Show your working steps, including truth tables if any.

(10 Marks)

Table 4

Select		Output
S_1	S_0	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

3. (a) Draw the gate level circuit of a positive-level trigger T flip-flop based on the cross-coupled NOR latch. Show the characteristic table and write the characteristic equation for the T flip-flop.

(8 Marks)

(b) You are required to design a sequence detector based on Mealy model state machine to detect an input sequence of 1101, with overlapping of input '1' allowed, using D flip-flops and necessary logic gates. Execute your design by answering the following questions:

(i) Draw the state diagram.

(ii) Derive the transition/output table of the state diagram.

(iii) Tabulate the excitation table.

(iv) Plot the K-maps and write the characteristic equations.

(14 Marks)

Note: Question No. 3 continues page 4.

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(c) Briefly discuss the possible problem for a flip-flop if the set-up time margin is negative and state one possible situation to cause the set-up time margin to be negative.

(3 marks)

4. (a) Determine how many address lines and data lines are needed for each of the following memory modules:

(i) 16 M × 32

(ii) 256 K × 5

(iii) 24 × 9

(6 marks)

(b) Figure 1 shows a special D-latch circuit with A as the input and Q_2 as the output, and the timing diagram for the input A and clock signal CLK.

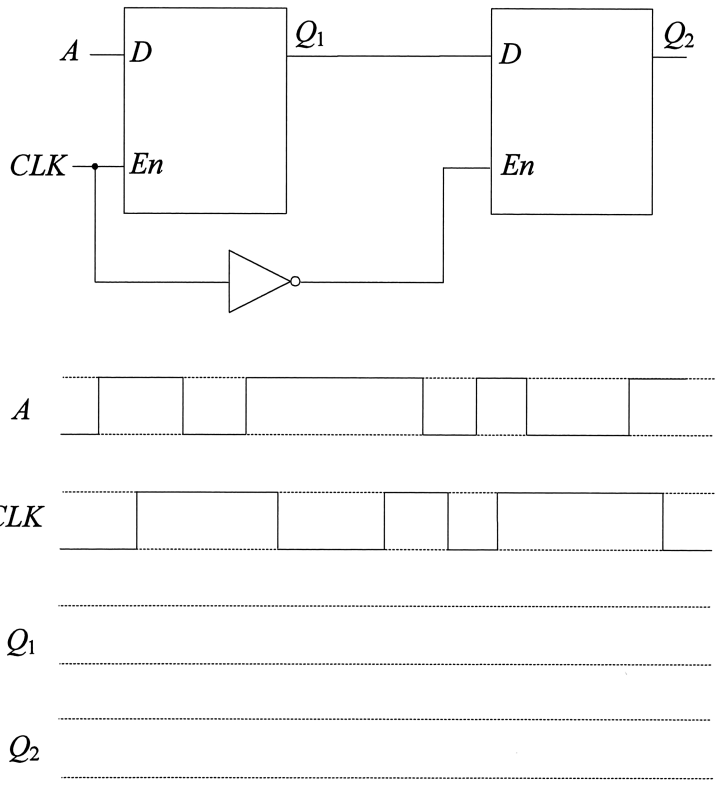


Figure 1

Note: Question No. 4 continues page 5.

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(i) Redraw the timing diagram on your answer booklet and determine the output states, Q_1 and Q_2, over time for the circuit.

(ii) At what specific times of *CLK* signal does output Q_2 follows input *A*?

(iii) What is the difference in term of triggering behaviour between the circuit shown in Figure 1 and a normal D-latch?

(14 marks)

(c) Discuss and compare the differences between synchronous and asynchronous count-up counters in terms of circuit area, power dissipation and design methodology.

(5 marks)

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.

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